MAE 155A - BCR

Team 2 - Pumpkin Patch

Blended Wing Body

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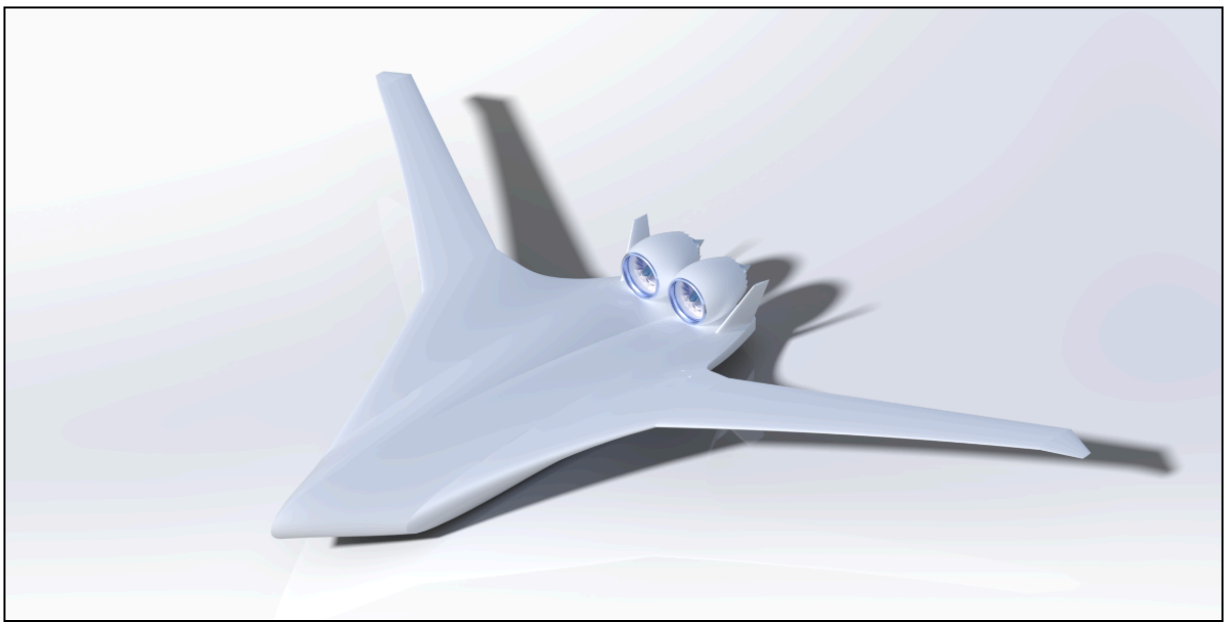


1. Executive Summary

1.1 Background and Motivation

Commercial aviation currently accounts for 2.4% of global CO2 emissions, contributing to irreversible climate change while simultaneously contributing to the operation cost for aviation companies [R1]. Thus, there is substantial demand for hyper-fuel efficient aircraft. Conventional tube-and-wing aircraft, like the Boeing Dreamliner 787-8, offer little improvements in aerodynamic efficiency due to the limitations set by their traditional design configuration. As such, there is a significant opportunity for dynamic improvements in fuel efficiency, motivating our implementation of a Blended Wing Body (BWB) aircraft design. For BWB aircraft, the fuselage becomes a central lifting body due to its exterior shape as an airfoil, resulting in significant contributions to overall aerodynamic efficiency, surpassing current conventional fuselage designs. Thus, perfecting a blended-wing-body design, and its inherent complications, will surely be the next big hurdle to overcome in the modern aviation world.

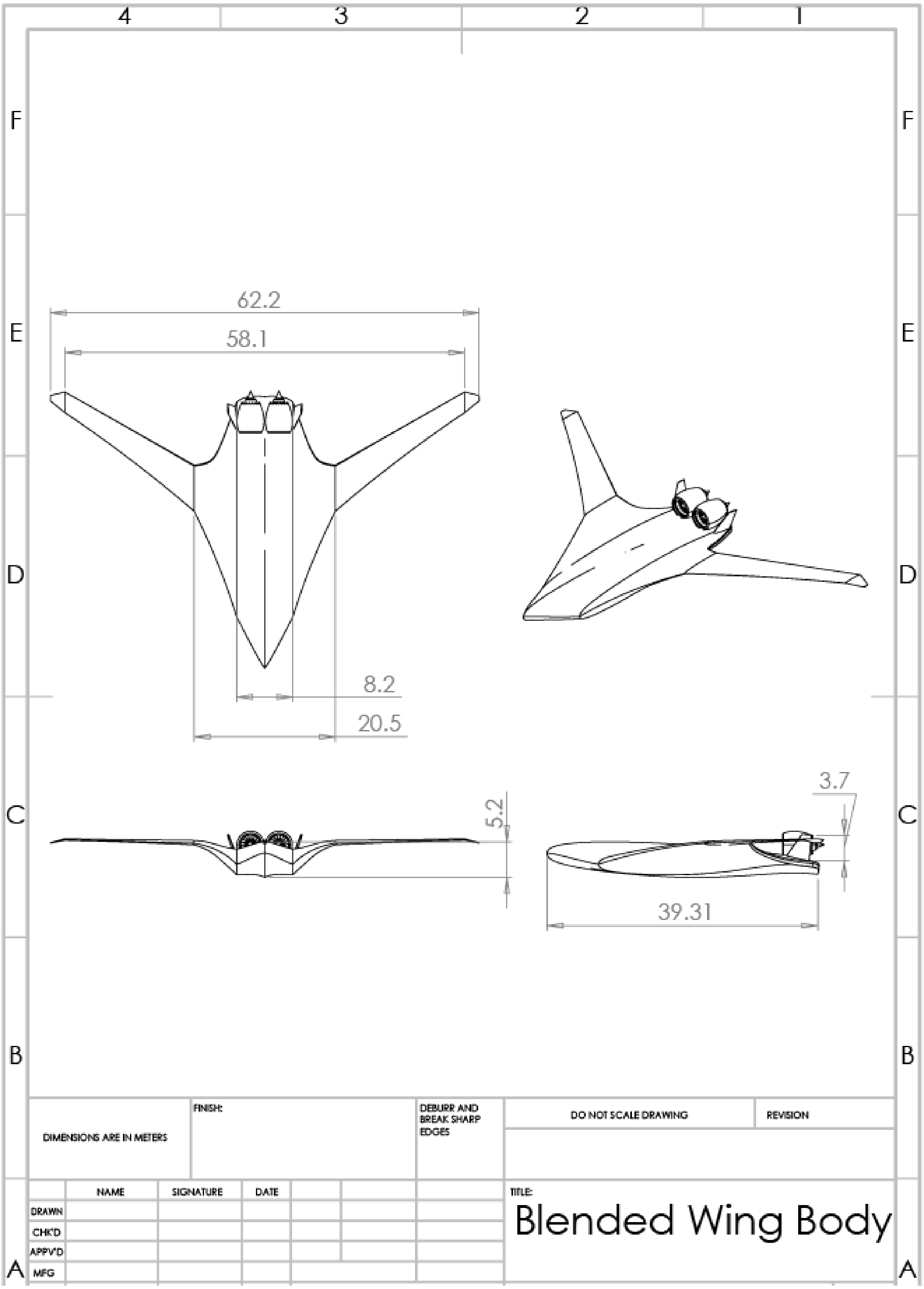
As a solution to this challenge, we propose a more efficient BWB commercial aircraft prototype with a maximum range of 7,400 nautical miles, a cruise speed of Mach 0.85, and a 250 passenger capacity. Compared to current prototypes, we intend to maximize our propulsive efficiency by harnessing the benefits of boundary layer ingestion through aft-mounted, smoothly integrated, high-bypass turbofan engines. Additionally, to combat inherent instability issues present in flying wing designs, twin-mounted inclined vertical stabilizers will provide adequate yaw control and stability throughout our mission profile. Finally, by taking advantage of new research in folding wing designs, we intend to minimize our fuel consumption while seamlessly integrating into existing infrastructure - a common negative of current prototypes. Therefore, we aim to target long-haul routes currently serviced by full-service carriers (FSCs) to best capitalize on the deployment of more fuel efficient aircraft.

1.2 Table of Main Aircraft Parameters

Range [nmi]	Cruise Mach	Altitude [ft]	Capacity [pax]	Wingspan [m]	AR [m]	S_{ref} [m]	MTOW [N]	Engine Type	Cruise L/D
7400	0.85	40,000	250	62.2	4.92	677	1,672,200	GEnX-1B	23.42

Table 1.1: Table of key aircraft specifications of our proposed BWB aircraft prototype

1.3 Technical Drawings



2. Introduction

2.1 Market Analysis and Design Problem

Fuel-efficient widebodies like the Boeing 787 and Airbus A350 currently dominate the global long-haul commercial aviation market. Despite constant incremental improvements in aerodynamic efficiency and development of higher strength-to-weight materials, conventional designs are ultimately restricted by their tube-and-wing architecture. Global initiatives for more sustainable air travel are incentivizing airlines to pursue hyper-fuel efficient aircraft that will reduce CO_2 emissions and increase profitability amidst rising fuel costs. The main design challenge is not only to construct a hyper-efficient next-generation widebody aircraft, but also to seamlessly integrate it with current airport infrastructure such that airlines can efficiently start implementing these aircraft into their global networks.

2.3 Reference Aircraft

Reference aircraft provide benchmarks for performance, passenger comfort, operational feasibility and manufacturability. The Boeing 787-8 was selected as the primary benchmark for existing aircraft due to its similar operating range and speed and its comparable passenger capacity. JetZero's Z4 was selected as the primary benchmark for aircraft concepts as it aims to fulfill a similar role to our aircraft concept and thus we can use it as a performance benchmark to evaluate our design against. Using both of these aircraft as references can help inform better key design choices like operating speed, structural layout as well as control design and propulsion integration.

2.4 Critical Technologies

To realize the blended-wing-body design, several technologies are essential. Lifting body aerodynamics allow the fuselage to generate a substantial portion of lift, while reducing drag and improving aerodynamic efficiency. The airframe is designed using structural composites, which provide high strength-to-weight properties, and allow for large and non-circular pressurized cabin volumes. Aft-mounted high-bypass turbofan engines are critical for achieving optimal propulsion performance, with significant benefits from boundary layer ingestion and reduced nacelle drag.

2.5. Concept of Operations

Our aircraft concept is designed to operate long-haul routes up to a total flight distance of 7400 nautical miles. The primary route outlined as our maximum flight distance is Los Angeles (LAX) to Dubai (DXB). Serving between major airport hubs, FSC airlines can operate our aircraft for high-demand routes to offer passengers premium comfort flights with our improved amenities and spacing. Alternatively, airlines can operate our aircraft between medium-demand routes to increase profitability and make previously unprofitable routes now profitable. Our aircraft concept is designed to cruise at 40,000ft at Mach 0.85 to optimize fuel efficiency and range. To maintain operational flexibility the design fits within Code E/V gates, allowing the design to fit within existing airport infrastructure to maximize the efficiency of fleet integration for airlines.

3. Initial Sizing

3.1 Initial Weight Estimation

To estimate the initial weights of our aircraft, we used empirical data for similar class aircraft to compute the payload, empty, and energy storage weight fractions as presented in Raymer section 3.3. Although the Jet Transport class of aircraft corresponds to a typical tube and wing style aircraft, we chose to use it as a baseline for predicting our BWB performance since the class contains the majority of our competing aircraft designs. Additionally, we calculated the fuel weight fraction estimates such that they fulfill the LAX to DXB mission profile as this is one of the longer flight routes we aim to cover. Fuel fraction estimates were optimized to minimize the penalties associated with carrying additional reserve weight across long distances. By modifying John T. Hwang's gross weight Marimo notebook to include an additional penalty for 30 minutes of loiter and a 45 minute potential diversion, we were able to compute the weight fractions shown below in table 3.1 using the inputs specified in appendix A1.

3.2 Wing and Engine Sizing

To determine an initial preliminary estimation for our wing loading $\frac{W}{S}$ and thrust to weight $\frac{T}{W}$, we implemented a 5% safety margin from all expected performance constraints. To perform the constraint analysis, a carpet plot was utilized to compare our selected wing loading and thrust-to-weight value to stall, climb, maneuver (approach and cruise), takeoff, landing, and ceiling constraints. John T. Hwang's sizing carpet plot Marimo notebook was utilized to generate the carpet plot shown in Figure 3.1, with corresponding input parameters displayed in appendix A2. The selected wing loading and thrust to weight ratio are also shown in table 3.1 below.

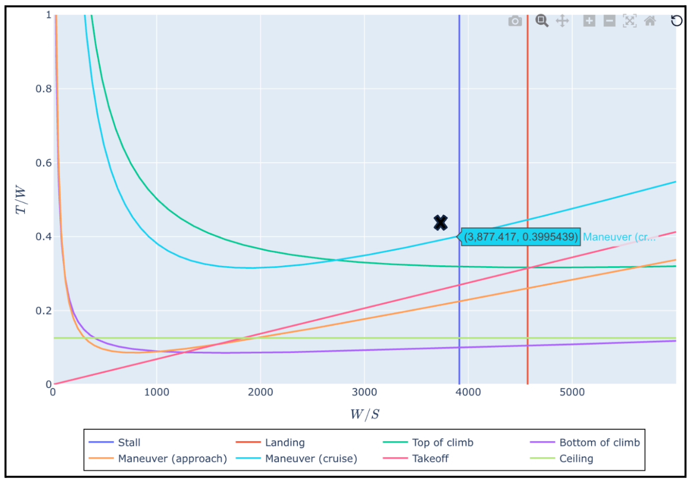
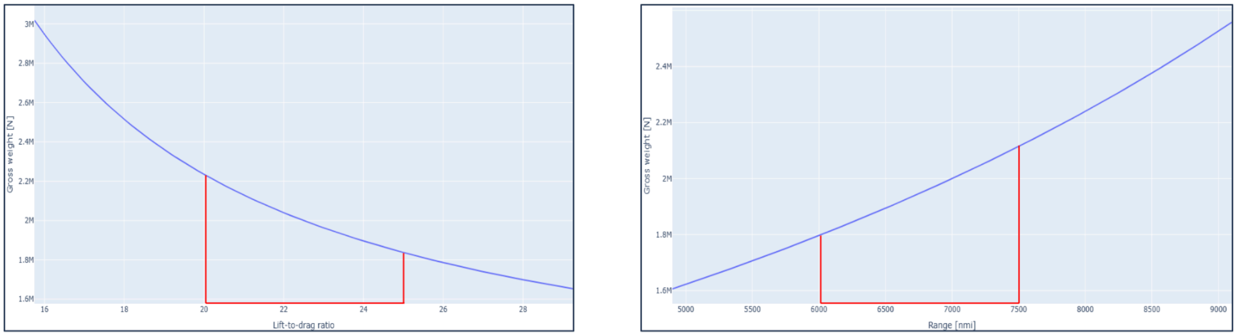


Figure 3.1: Thrust-to-weight ratio ($\frac{T}{W}$) versus wing loading ($\frac{W}{S}$) diagram illustrating key aircraft performance constraints. The black x indicates a 5% margin from the maneuver (cruise) and stall constraints.

Gross Weight [N]	Payload Weight Fraction	Empty Weight Fraction	Energy Storage Fraction	W/S [N/m^2, lb/ft^2]	T/W
1,707,300	.23	0.41	0.36	3683 , 80.5	0.42

Table 3.1: Initial gross weight estimation W_g and corresponding payload ($\frac{W_p}{W_g}$), empty weight ($\frac{W_e}{W_g}$), and energy storage ($\frac{W_s}{W_g}$) weight fractions, as well as initial wing loading and thrust to weight ratios.

3.3 Sensitivity Analysis



Parameter sweep: Gross weight vs L/D

Parameter sweep: Gross weight vs Range

Figure 3.2: Parameter Sweep (Gross weight VS L/D) and (Gross weight vs Range)

As pictured in Figure 3.2 above, as L/D increases, gross weight decreases nonlinearly as a result of improved aerodynamics that in turn reduces fuel weight requirements. Conversely, as range increases gross weight rises due to additional fuel mass required to complete mission requirements. This also increases structural weight effects required for additional fuel storage. The highlighted red regions represent the design parameters evaluated in order to complete our mission objectives.

4. Concept

4.1 Passenger and Cargo Configuration

Our aircraft concept adopts a triple-aisle main cabin configuration optimized for high-capacity long-range operability while maximizing passenger comfort. The first class cabin is configured in the triangular front section of the airframe, with 20 first-class seats designed to rival the best first-class suites currently available in competing aircraft. The unconventional first class cabin space also allows for premium lavatories and additional space that can accommodate a shower or meeting room depending on specific airline requirements. The main cabin, composed of 3 rows of business seats and 16 rows of economy seats incorporates a 2-4-4-2 seating arrangement to prioritize passenger comfort and maintain safety regulations, particularly for emergency situations. There are 3 main, floor-level exits equidistantly spaced on either side of the fuselage to ensure that there are ample routes for passengers to disperse during emergency situations. The cockpit is located at the front of the plane, and is designed to accommodate 2 pilots. See section 4.4 for additional information on cockpit configuration. The cabin crew have designated quarters underneath the first class section. We designed the fuselage so that the cargo section is housed below and to either side of the main cabin. The cargo section uses the LD-3 cargo containers as these provide the best fit to our widebody design. A visualization can be seen on the right in Figure 4.1.

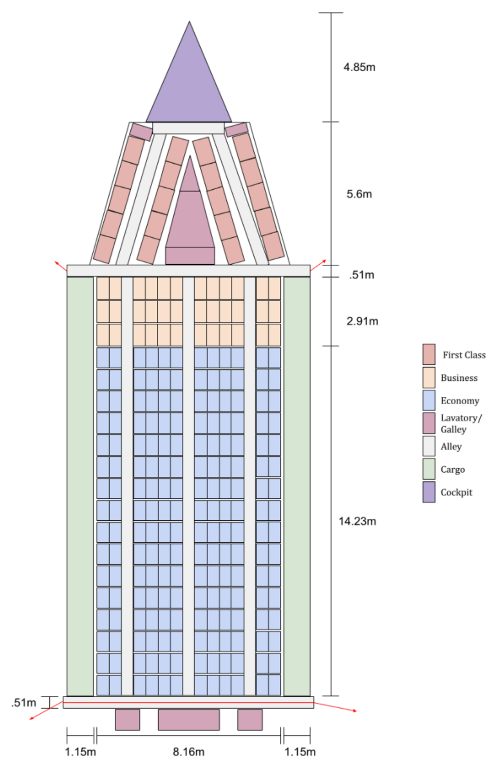


Figure 4.1: Diagram of aircraft cabin configuration including cockpit, lavatories, seating, aisles, emergency exits, engines, control surfaces, fuel tanks and vertical stabilizer CG.

4.2 Fuselage Design

Due to the design of our passenger and cargo areas, our BWB necessitated an elliptical fuselage instead of a traditional cylindrical fuselage. As such, the pressurization loads which are typically handled by a more stress optimal circular cross-section must be accounted for with careful consideration in our design. Our proposed design features unique structural elements to support a single unified cabin bay, allowing for a wider and more oval cabin section due to our wide body. In this design, the pressurization loads are combated by implementing new technology in advanced composites called Pultruded Rod Stitched Efficient Unitized Structure [R12]. Pultruded rods are used to line the inside of the skin panels to stiffen the body against buckling and carry the axial tension from pressurization. Thus, bending moments from internal pressure and distributed aerodynamic loads will have minimal impact on our BWB's structural integrity. Additionally, stiffened shell panels are utilized to prevent structure collapse during yielding.

4.3 Nose and Landing Gear Design

For the landing gear configuration, a multi-bogey landing gear system will be integrated into our BWB aircraft. This configuration is selected to accommodate the high max takeoff weight (MTOW) and aligns structurally with the nature of our wide body, ensuring reliable load distribution during commercial operations. The landing gear will consist of three retractable struts with four wheels each and the nose gear will have two wheels. The main strut will be in line with the centerline of the aircraft, there will be two adjacent struts towards the wings. The nose gear will also be in line with the aircraft's centerline. Based on a CG excursion analysis, the aircraft center of gravity varies between 21.83 - 23.16 m from the nose as fuel is consumed during flight. Applying the 15° criterion, the minimum main landing gear placement was determined to be 24.18 from the nose, ensuring adequate rotation capability. As such, the two side struts adjacent to the main landing gear will be 3.2 m offset from the aircraft centerline and 22.7 m from the nose. Therefore, the nose gear was determined to be most suitable at 8.217 m. In this configuration, the static configuration load fraction will be a maximum of 14.6% across all loading cases, which lies between the recommended range of 5%-20% for transport aviation.

To determine the required main landing gear height, a potential rotational strike zone was evaluated to be the underside of the tail based on our preliminary geometry. A maximum rotation angle of 12° was selected to represent take off pitch conditions similar to modern transport aircraft. Additionally, a clearance margin of 0.5 m was applied to account for strut compression or runway irregularities to ensure safety. Using these assumptions the main landing hear height was calculated to be about 3.3 meters.

Main wheel sizing was performed using the transport category empirical correlation [R2]. The main landing gear supports 85.4% of the aircraft weight based on the maximum nose gear load fraction of 14.6%. Applying transport conditions produces an estimated tire diameter of 0.94 m and a width of .31 m.

4.4 Cockpit Visibility

The cockpit utilizes a four pane glass window to enhance pilot visibility. The cockpit is designed so that the pilots are aware of their surroundings during takeoff, cruise, and landing conditions. The pilots are able to see 10 degrees below the horizon during landing and 5 degrees below the horizon during initial climb.

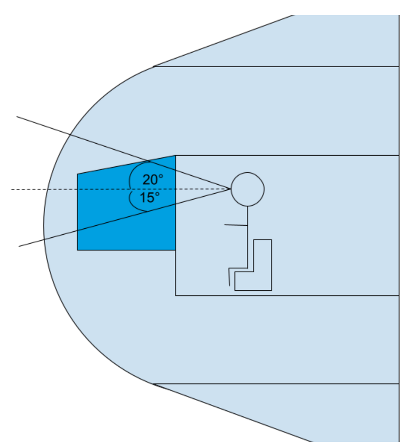
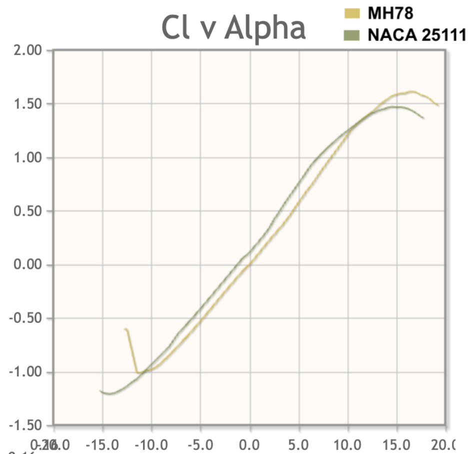


Figure 4.2: Side view schematic of the cockpit outlining visibility angles for a 5'10'' pilot.

5. Aerodynamics

5.1 Airfoil Selection

The aircraft was divided into two main lifting regions to guide aerodynamic design. The center body houses passengers, cargo, and primary structural elements, while the outer wing focuses on aerodynamic efficiency and supporting control surfaces. Airfoil selection started with the center body because it has the largest surface area and most significant contribution to the aircraft's maximum C_L. The NACA 25111 was evaluated to produce the highest lift coefficient among airfoils that were feasible to integrate into BWB designs [R3]. A byproduct of this lift, however, was a negative moment coefficient that would affect the aircraft's stability (Figure 5.1b). Outer wing design was driven by finding airfoils that produced a positive moment to counteract the center body. Similar to NACA 25111, Airfoil MH78 was selected for the outer wing due to its ease of integration (Figure 5.1b) [R3].



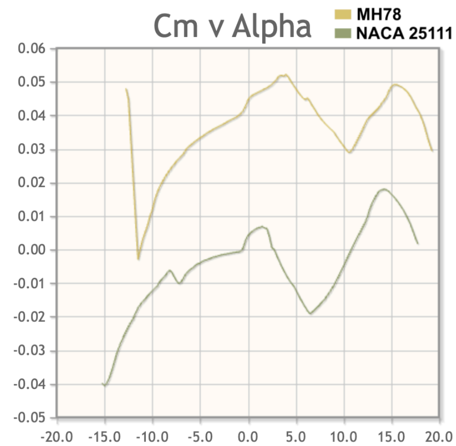


Figure 5.1: MH78 and NACA 25111 changes in (a) Lift Coefficient and (b) Moment Coefficient as the angle of attack (degrees) is varied from -20° to +20°.

5.2 Wing Design

Initial sizing of the fuselage body stemmed from the cabin layout detailed in section 4.1. An insulation of 2% around the interior cabin space with an additional buffer of 1.75 inches was used. The rear of the pressurized cabin was assumed to be at 70% of the root chord in order to handle bending and shear loads from the outer wing section [R9]. Following calculations outlined in Appendix A3, the total fuselage length was found to be 39 meters. Accounting for insulation requirements on the cabin width, as well as space added from adjacent cargo, the fuselage width was calculated to be 10.71 meters. Cabin height was sized using the fuselage length and thickness-to-chord ratio of the NACA 25111 airfoil.

Outer wing design was driven by compliance with Code E gate restrictions following market analysis outlined in section 2.1. The wings were sized to meet the maximum allowable wingspan of 65 using our fuselage dimensions. Following span selection, the sweep angle was determined to support efficient transonic cruise performance and delay the onset of compressibility effects like wave drag.

Finally, the taper ratio of the airfoils was chosen using wing span, sweep angles, and CFD analysis from reference BWB case studies. Notably, BWB designs feature higher taper ratios compared to traditional aircraft in order to balance structural efficiency, optimize span-wise lift distribution, and minimize induced drag. These airfoil design parameters are summarized in Appendix A6.

5.3 Lift and Drag Analysis

Lift and drag analysis was conducted using empirical formulas to translate two-dimensional airfoil coefficients to spanwise aircraft coefficients. With a Reynolds number on the order of 10e+8 and a transonic Mach number of 0.85, the lift coefficient was calculated using equations detailed in the aerodynamics section of Raymer's Textbook [R2] and airfoil geometry parameters in Table 5.1. Total drag was aggregated from contributions of wave, parasite, and lift-induced drag, and computed using similar empirical relations derived in Raymer [R2]. The spanwise aerodynamic parameters were derived using a sectional aerodynamic model that sums the contributions from the three different airfoil regions by constructing individual sectional lift curves, computing a total lift coefficient, and then forming the base drag coefficient as a sum of individual section's parasitic drag and induced drag contributions.

The spanwise aerodynamic coefficients are in section 5.4, Table 5.2, along with graphs detailing the drag polar, lift coefficient, total drag rise and overall aerodynamic efficiency , as seen in Figure 5.2 below.

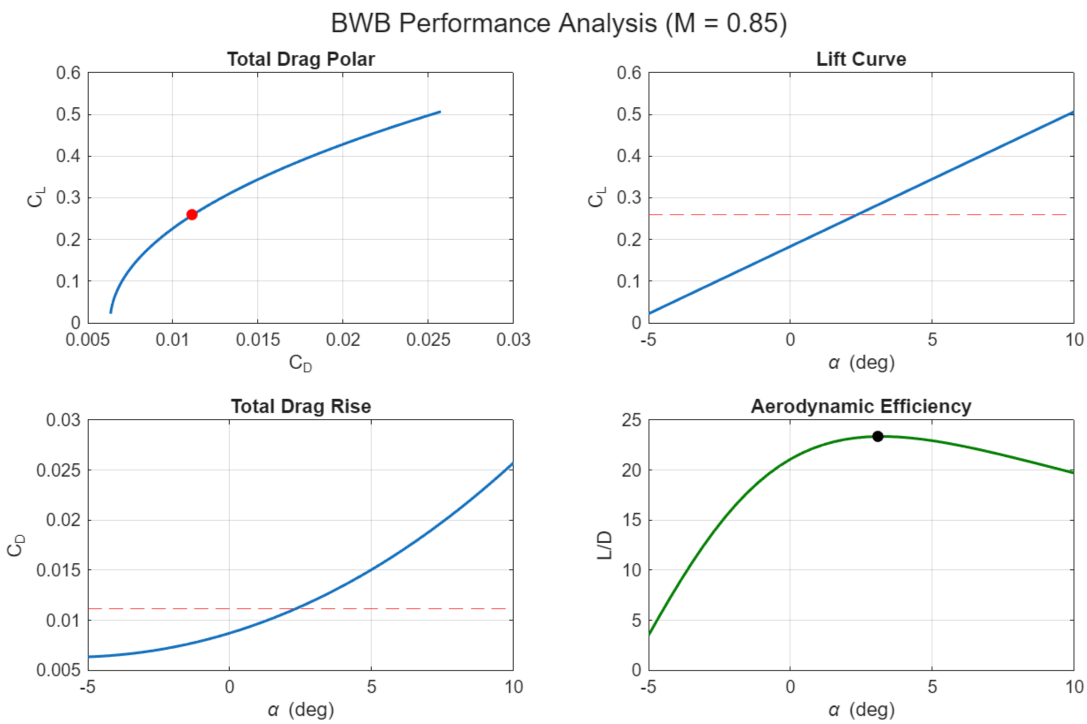


Figure 5.2: From top left to bottom right, the figure shows the Total Drag Polar (C_L vs. C_D) as a solid blue line with red marker indicating cruise conditions, Lift and Drag Curves vs. AoA (degrees) in solid blue with a dotted red line indicating cruise conditions, and the L/D vs. AoA (degrees) in solid green line with a solid black marker showing max aerodynamic efficiency.

5.4 Table of Key Aerodynamic Parameters During Cruise

C_L	AoA (°)	$C_{D,0}$	$C_{D,i}$	$C_{D,w}$	C_D	$\frac{L}{D}$	$\frac{dC_L}{d\alpha}(\pi)$
0.2604	2.3077	0.0063	0.0047	0.0002	0.0111	23.421	1.8507

Table 5.2: Aerodynamic parameters at a 2.31° angle of attack during cruise.

6. Propulsion

6.1 System Architecture and Integration

Our aircraft concept employs a twin-engine propulsion configuration, consisting of two GEnX-1B high-bypass turbofan engines mounted on the aft upper surface of the airframe. This system architecture was selected to maximize propulsion efficiency and integration feasibility while using currently available propulsion technologies. Mounting the engines at the rear will reduce any aerodynamic interference with primary lifting surfaces, and enable the ingestion of the low-energy boundary layer flow moving along the upper surface of the airframe. By smoothly integrating the nacelles into the airframe, boundary layer ingestion is maximized, leading to improved propulsive efficiency through reduced nacelle drag and lower free-stream velocity. Additionally, upper-surface engine mounting will provide acoustic shielding by the airframe to help reduce the aircraft's overall acoustic signature while also minimizing ground clearance and foreign object damage risks during operations. To structurally integrate the engines into the airframe, internal reinforced spars will allow for efficient management of engine loads.

6.2 Component Selection and Sizing

The GEnX-1B turbofan engine is selected due to its high-bypass ratio and proven long-haul operability and reliability. This engine provides a maximum takeoff thrust of 330kN, which satisfies all of our aircraft's maneuver requirements including takeoff, climb, loitering, and landing while upholding engine-out safety margins. The high-bypass ratio leads to a lower overall specific fuel consumption, helping to reduce overall fuel consumption and maximize total efficiency.

6.3 Table of Key Propulsion Parameters

TSFC $(\frac{1}{s})$	Bypass Ratio	Weight (N)	Thrust (N)	Mass Flow $(\frac{Kg}{s})$	Inlet Diameter (m)	Pressure Ratio	Engine Count
1.41e-05	9.1	62,264	330,000	1160	2.82	43.7	2

Table 6.1: Important propulsion system parameters for our BWB aircraft with a GEnX-1B engine [R13].

7. Weights and Balance

7.1 Weight Breakdown

The weight calculations were conducted by referencing Raymer's Chapter 15, section 15.3.3 equations for general aviation weight breakdown [R2]. Using our initial guess for gross weight developed in section 3.1, Raymer's equations were adapted to determine the breakdown of our MTOW through an iterative method. As a result, our final MTOW was found to be 1.67e+06 newtons. Notably, even when varying our initial guess of MTOW by several orders of magnitude, our final MTOW was still determined to be 1.67e+06 newtons, which is within margin of the MTOW obtained during initial sizing in section 3.1. The total weight breakdown is shown in appendix A4 and Figure 7.1.

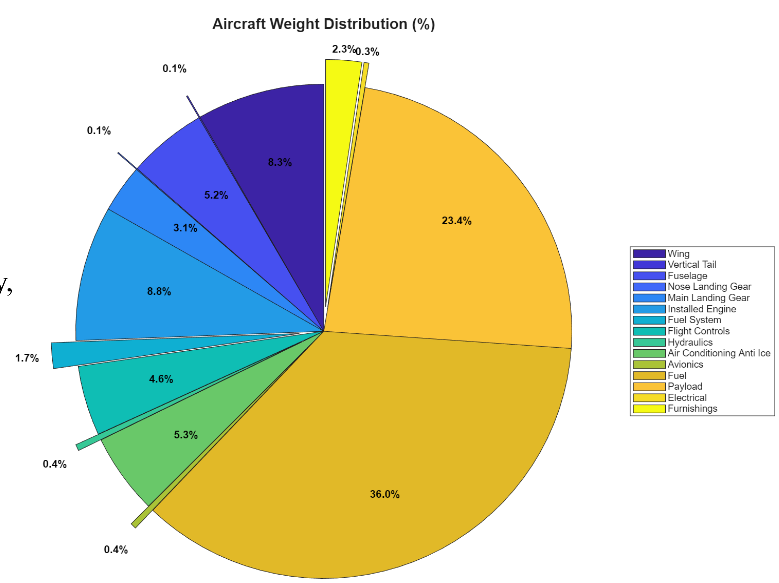
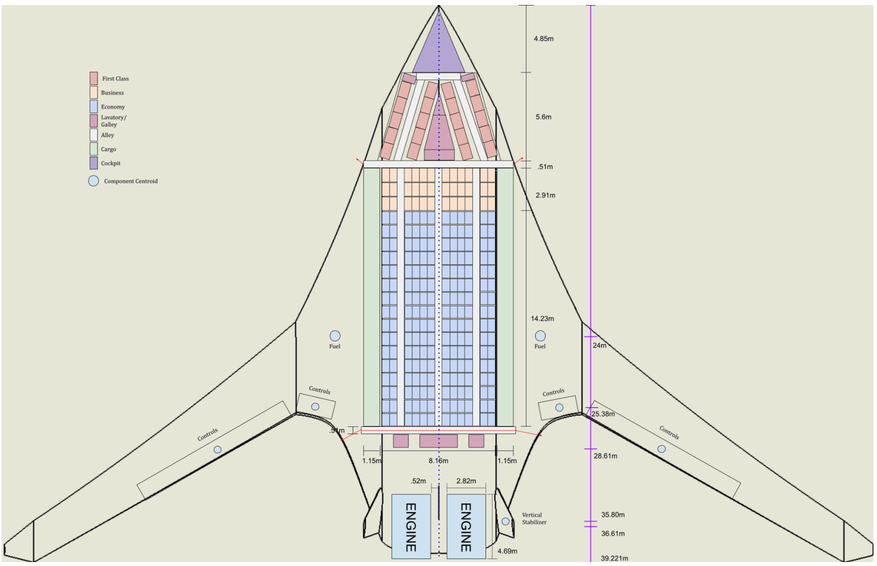


Figure 7.1: Pie chart breaking down the component specific weight contribution for our BWB design.

7.2 Stability Breakdown

To evaluate the stability of our aircraft, the neutral point (Np), center of gravity (Xcg), and static margin (SM) were computed across the course of our design mission. The process in determining each parameter is detailed below.

The Neutral Point was found by averaging the results of two methods: (1) an empirical evaluation based on sectional airfoil properties and planform geometry, and (2) a direct computation from OpenVSP at our nominal cruise condition.



The longitudinal Xcg of the aircraft was determined by aggregating the individual contributions of each major weight component, with their respective Xcg locations shown in Figure 7.2. A CG excursion was evaluated across our design mission by determining the quantity of fuel remaining after each mission phase and then recalculating the total aircraft Xcg. Pictured in Figure 7.2, the Xcg of each major component can be seen. The other two Xcgs are assumed to be close to zero, and are therefore not evaluated for the purposes of the stability analysis.

Figure 7.2: CG location of each major component on our baseline BWB. Distances shown are measured from the nose backward.

The SM was computed as a percentage of the mean aerodynamic cord, and was found by taking the Np minus the Xcg over the MAC. Notably, as the airplane goes through the different mission phases, the SM of the aircraft increases due to the decrease in fuel weight located aft of the aircraft Xcg. A plot of the static margin during the different mission phases is included in Appendix A7.

7.3 Key Stability Parameters

Parameter	Np	Xcg	SM
Meters, % MAC	24.5 , 108	22.5 , 99.2	1.91 , 8.40

Table 7.3: Key longitudinal stability parameters for our baseline BWB configuration at MTOW.

8. Structures and Materials

8.1 Overall Structural Layout

Material Selection for our blended wing body aircraft was driven primarily by our objective of achieving ultra-fuel efficiency. This involves structural weight reduction without sacrificing durability and manufacturability for commercial service. The primary lifting structures, including the centerbody and wings, will be carbon fiber reinforced polymer (CFRP), while keel beams and I-beam supporting structure will be High Modulus Carbon fiber. This is primarily due to carbon fiber's high specific strength and stiffness, enabling significant reduction in empty weight compared to metallic structures. Aluminium was selected for the wing and vertical tail ribs for its ease of manufacturing. Titanium was chosen for design in high-temperature and high-load environments, such as engine mounts as well as wing-body transition sections. Steel is reserved for landing gear components where fatigue resistance and strength are essential. Metallic reinforcements (Titanium, Aluminum) are incorporated in impact-prone areas such as leading edges to improve damage tolerance. This combination of materials minimizes structural weight, supporting overall fuel efficiency while still emphasizing safety and longevity in our design. Figure 8.1 below shows the bubble structural layout of our BWB model.

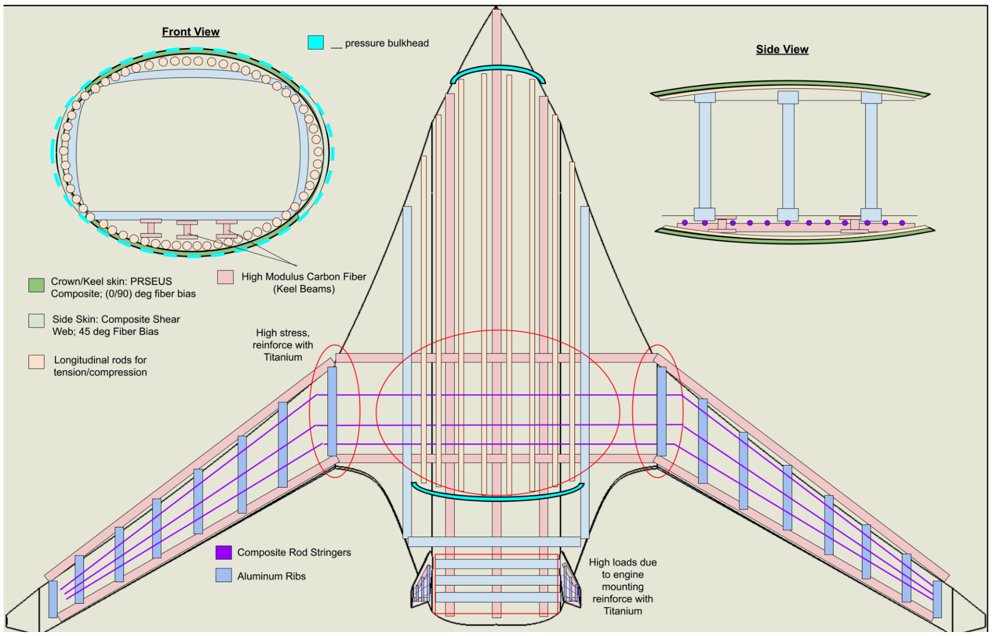
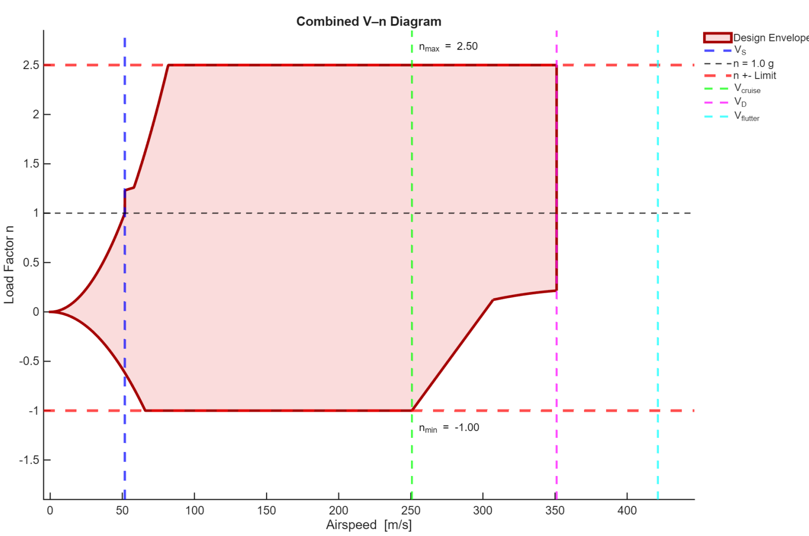


Figure 8.1: Preliminary structural layout of our baseline BWB configuration. Pictured on the top left is the front view of the airplane taken from the nose, the top right shows the side view of the central body, and the middle view is a top down look of the airplane.

8.3 V-n Diagram



The V-n Diagram is used in this study to define the structural load envelope for the blended wing body aircraft across the operating speed range. It provides the basis for evaluating maneuver and gust loads and is used to support preliminary sizing. The resulting envelope is consistent with transport class limits.

Figure 8.2 V-n diagram showcasing the load factor, n, versus airspeed for various mission phases including cruise,stall, dive, and flutter.

9. Conclusion

	Gross Weight [N]	Wp/Wg	We/Wg	Ws/Wg	W/S [N/m^2]	T/W
Initial	1,707,300	.23	0.41	0.36	3683	0.42
Final	1,672,200	.23	0.41	0.36	2470.35	0.39

Table 9.1: Initial sizing parameters compared to our final BCR sizing parameters.

Table 9.1 shows strong agreement between the initial and refined sizing with only a ~2% reduction in gross weight after iterative analysis. The payload, empty, and energy storage fractions remained unchanged. This confirms that mission requirements were preserved over the design process.

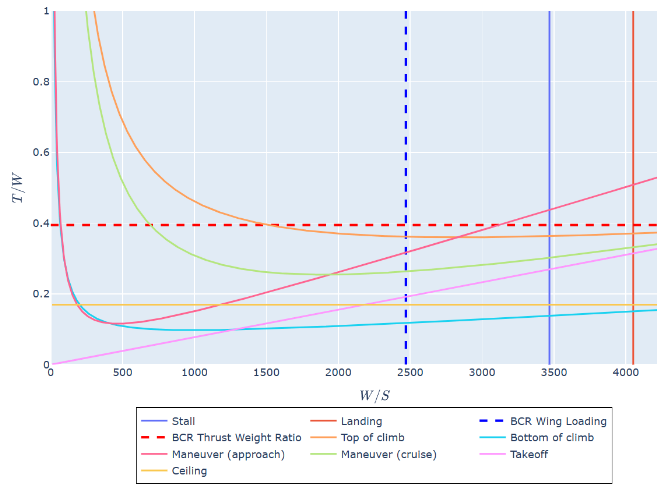


Figure 9.2 T/W ratio versus W/S carpet plot illustrating aircraft performance constraints. The dashed red and blue lines indicate the T/W and W/S respectively for our BCR configuration.

As illustrated by Figure 9.2, the final design point lies comfortably within the allowable bounded region of our mission profile. Overall the constraint diagram validates the current configuration of our aircraft.

Our conceptual design of a Blended Wing Body aircraft is motivated by the prevailing demand for hyper fuel-efficient commercial aircraft which cannot be met by the traditional tube and wing design. Our proposed BWB aircraft, designed for a range of 7400 nautical miles, a flight speed of Mach 0.85, and a passenger capacity of 250, has been designed using a Multidisciplinary Design Optimization (MDO) method in which various parameters governing crucial flight parameters such as aerodynamics, weights, structures, and propulsion were varied to obtain the most efficient BWB design. The iterative process yielded multi-faceted improvements in various flight performance areas. As Table 9.1 shows, the refined sizing yielded a 2% reduction in gross weight, with the payload, empty, and energy storage fractions remaining unchanged. This confirms that mission requirements were preserved over the design process, while weight was optimized. Additionally, Figure 9.2 depicts a substantial reduction of wing loading, from ~3600N/m^2 to ~2500 N/m^2, ensuring better stall and landing margins for our aircraft and overall improvements in aerodynamics. Throughout the design process, special attention was paid to the inherent stability and structural issues evident in previous commercial BWB designs, resulting in the implementation of twin-mounted inclined vertical stabilizers and a novel fuselage design. Attention was also paid to developing an aircraft which can be integrated into existing airport infrastructure, resulting in the incorporation of foldable wings. With this design, our proposed aircraft will allow commercial airlines to enter a revolutionary new era of accessible hyper fuel-efficient air travel.

10. References

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11. Appendices

A1: Initial Sizing - Weight Fraction Input Parameters

Sizing parameters

Aircraft type: Jet-propelled

Empty weight fraction regression parameters: Coefficient, a: 0.97 | Exponent, b: -0.06

Payload weight, W_p [N]: 392000 | Range, R [m]: 1.3705e7 | Cruise speed, V [m/s]: 250.7

Lift-to-drag ratio, L/D: 22.5 | TSFC, c_T [kg/N/s]: 1.415e-5 | BSFC, c_P [kg*s/W]: 5e-8

Efficiency, shaft to thrust power, η_s: 0.9 | Efficiency, battery to thrust power, η_b: 0.9

Battery specific energy, e_b [J/kg]: 7e5 | Battery specific energy, e_b [J/kg]: 4.4e7

Loiter Time, t_l [min]: 30 | Additional Cruise Time, t_c [min]: 45

A2: Initial Sizing - Wing Loading and Thrust To Weight Input Parameters

Sizing parameters

Cruise speed, V_{cruise} [m/s]: 250.7 | Climb speed, V_{climb} [m/s]: 73.56 | Stall speed, V_{stall} [m/s]: 61.3

Takeoff speed, V_{takeoff} [m/s]: 67.43 | Approach speed, V_{approach} [m/s]: 79.7

Cruise altitude, h_{cruise} [m]: 12192 | Parasitic drag coeff., C_{D0}: 0.0078

Max. lift coeff., $C_{L\text{max}}$: 1.7 | Takeoff lift coeff., $C_{L\text{takeoff}}$: 1.405

Maneuver load factor, n: 2.5 | Climb angle, γ [deg]: 2.7

Oswald efficiency factor, e: 0.85 | Aspect ratio, AR: 12

Thrust Mach lapse, k ($T/T_{sl} = 1 - k \cdot M$): 0.4 | Thrust density lapse, m ($T/T_{sl} = (\rho/\rho_{sl})^m$): 0.7

Takeoff density altitude, h_{takeoff} [m]: 1609. | Landing density altitude, h_{landing} [m]: 1609.

Takeoff parameter coefficient, k_{takeoff} [m^3/N]: 0.255 | Landing parameter coefficient, k_{landing} [m^3/N]: 0.51

Landing dist., d_{landing} [m]: 1905 | Airborne dist., d_{airborne} [m]: 300. | Takeoff dist., d_{takeoff} [m]: 3352.8

A3: Initial Sizing - Interior and Exterior Sizing Calculations

Fuselage Width:
12 Passenger Seats: 54cm/seat → 6.48m
3 Aisles: 51cm/aisle → 1.53m
Cargo: 2.3m
R&L Wing Padding: 0.15m
<u>Interior Cabin Width</u>: 10.46m
Fuselage + Insulation: 2% interior width + 1.75in
<u>Exterior Cabin Width</u>: 10.71m
Chord Length:
Main Cabin: 17.14m
Cockpit length (Statistical): 4.5m
First Class: 6.2 m
Alley (Front + Back): 1.02m
Lavatory (Front + Back): 2m
<u>Interior Cabin Length:</u> 30.86m

Fuselage + Insulation: 2% interior width + 1.75in Estimated Exterior Length: 31.52m Assumption of 70-80% of total chord Chord Length: 39m **Fuselage Height:** Centre t/c: 11% Fuselage Height: 4.29m

A4: Weights and Sizing: weight totals and percentages

Weight Segment	Weight (Newtons)	Weight (%)
Wing	1.39e+05	8.32
Vertical Tail	1.33e+03	0.079
Fuselage	8.65e+04	5.17
Main Landing Gear	5.24e+04	3.13
Nose Landing Gear	1.10e+03	0.066
Installed Engines	1.48e+05	8.83
Fuel System	2.80e+04	1.67
Flight Controls	7.63e+04	4.56
Hydraulics	6.90e+03	0.41
Avionics	6.50e+03	0.39
Electrical	5.38e+03	0.32
Air Conditioning Anti Ice	8.80e+04	5.26
Furnishings	3.92e+04	2.34
Fuel	6.02e+05	36.00
Payload	3.92e+05	23.44
Total	1.67e+06	100.00

A5: Empirical Aerodynamic Relations

Wave Drag: $C_{D_W} = 20(M - M_{Cr})^4$; where M is the Mach Number, and M_{Cr} is the critical Mach Number.

$M_{DD} = \kappa_A - (\frac{t}{c}) - 0.1C_L$; where M_{DD} is the drag-divergence Mach Number, κ_A is the airfoil technology factor (0.95 for supercritical airfoils), $\frac{t}{c}$ is the thickness-to-chord ratio and C_L is the lift coefficient.

$$\frac{dC_{D_w}}{dM} = 0.1$$

A6: Geometric and Aerodynamic Parameters defining baseline BWB Configuration

Aspect Ratio	4.923
Total Wing Span (m)	62.2
Fuselage Height (m)	5.2
S_{wet}/S_{ref}	2.28
LE Sweep Center Airfoil (°)	62
LE Sweep Outer Airfoil (°)	38

NACA 25111 Center Airfoil t/c (%)	11
NACA 25111 Outboard Airfoil t/c (%)	10
NACA 25111 Center Chord Length (m)	39.31
NACA 25111 Outboard Chord Length (m)	30.8
MH78 Outboard Airfoil t/c (%)	8
MH78 Outboard Airfoil Chord Length (m)	9

On the left are parameters pertaining to the entire body, and on the right are airfoil-specific parameters.

A7: Displayed change in SM as a percentage of the MAC during our design route.